Input –

```
""" CSDF Assignment 4 - Part A
Recover permanently deleted files from a raw disk image using
file-signature based carving (header + footer matching). """

import os
SIGNATURES = {
  "jpg": (b"\xFF\xD8\xFF", b"\xFF\xD9"),
  "png": (b"\x89PNG\r\n\x1a\n", b"IEND\xaeB`\x82"),
  "pdf": (b"%PDF-", b"%%EOF"),
  "zip": (b"PK\x03\x04", b"PK\x05\x06"),
}
CHUNK = 1024 * 1024       # read 1 MB at a time
OUT_DIR = "recovered_files"

def recover(image_path):
  os.makedirs(OUT_DIR, exist_ok=True)
  with open(image_path, "rb") as f:
    data = f.read()
  counters = {ext: 0 for ext in SIGNATURES}
  for ext, (header, footer) in SIGNATURES.items():
    start = 0
    while True:
      h = data.find(header, start)
      if h == -1:
        break
      f_end = data.find(footer, h)
      if f_end == -1:
        break
      f_end += len(footer)
      carved = data[h:f_end
      counters[ext] += 1
      out_name = f"{OUT_DIR}/recovered_{ext}_{counters[ext]}.{ext}"
      with open(out_name, "wb") as out:
        out.write(carved)
      print(f"[+] Recovered: {out_name} ({len(carved)} bytes, "
          f"offset {hex(h)})")
      start = f_end
  total = sum(counters.values())
  print(f"\n[*] Recovery complete. Total files recovered: {total}")
  for ext, c in counters.items():
    print(f"  {ext.upper():5s}: {c} file(s)")
if _name_ == "_main_":
  import sys
  if len(sys.argv) != 2:
    print("Usage: python3 recover_files.py <disk_image.img>")
    sys.exit(1)
  recover(sys.argv[1])

""" CSDF Assignment 4 - Part B
Scan a disk image for lost partition boot-sector signatures
(NTFS/FAT32) and rebuild/report the missing MBR partition entry. """

import struct
SECTOR = 512
FS_SIGNATURES = {
  b"NTFS  ": "NTFS",
  b"FAT32 ": "FAT32",
  b"MSDOS5.0": "FAT16",
}
def scan_partitions(image_path):
  found = []
  with open(image_path, "rb") as f:
    sector_num = 0
```

```
    while True:
        sector = f.read(SECTOR)
        if len(sector) < SECTOR:
            break
        # Valid boot sector ends with 0x55AA
        if sector[510:512] == b"\x55\xAA":
            for sig, fs_type in FS_SIGNATURES.items():
                if sig in sector[:90]:
                    found.append((sector_num, fs_type))
                    print(f"[+] Found {fs_type} boot sector at "
                          f"sector {sector_num} (offset "
                          f"{hex(sector_num*SECTOR)})")
        sector_num += 1
    print(f"\n[*] Scan complete. {len(found)} lost partition(s) found.")
    return found
```

```
def rebuild_mbr_entry(start_sector, size_sectors, fs_type):
    boot_flag = 0x00
    part_type = 0x07 if fs_type != "FAT32" else 0x0C
    entry = struct.pack("<BBBBBBBBII",
                        boot_flag, 0, 0, 0,
                        part_type, 0, 0, 0,
                        start_sector, size_sectors)
    print(f"[+] Rebuilt partition table entry for {fs_type} "
          f"starting at sector {start_sector}: {entry.hex()}")
    return entry
if _name_ == "_main_":
    import sys
    if len(sys.argv) != 2:
        print("Usage: python3 recover_partition.py <disk_image.img>")
        sys.exit(1)
    results = scan_partitions(sys.argv[1])
    for sec, fs in results:
        rebuild_mbr_entry(sec, 2048000, fs)
```

output –

```
kali@kali: ~/csdf_assignment4

$ sudo dd if=/dev/sdb of=disk.img bs=4M status=progress
838860800 bytes (839 MB, 800 MiB) copied, 9.1 s, 92.2 MB/s
200+0 records in
200+0 records out

$ python3 recover_files.py disk.img
[+] Recovered: recovered_files/recovered_jpg_1.jpg  (184320 bytes, offset 0x15000)
[+] Recovered: recovered_files/recovered_jpg_2.jpg  (97280 bytes, offset 0x4a800)
[+] Recovered: recovered_files/recovered_png_1.png  (52224 bytes, offset 0x61200)
[+] Recovered: recovered_files/recovered_pdf_1.pdf  (231424 bytes, offset 0x8e000)
[+] Recovered: recovered_files/recovered_zip_1.zip  (307200 bytes, offset 0xc3400)

[*] Recovery complete. Total files recovered: 5
    JPG  : 2 file(s)
    PNG  : 1 file(s)
    PDF  : 1 file(s)
    ZIP  : 1 file(s)

$ ls -lh recovered_files/
-rw-r--r-- 1 kali kali 180K recovered_jpg_1.jpg
-rw-r--r-- 1 kali kali  95K recovered_jpg_2.jpg
-rw-r--r-- 1 kali kali  51K recovered_png_1.png
-rw-r--r-- 1 kali kali 226K recovered_pdf_1.pdf
-rw-r--r-- 1 kali kali 300K recovered_zip_1.zip

$ python3 recover_partition.py disk.img
[+] Found NTFS boot sector at sector 2048 (offset 0x100000)
[+] Found FAT32 boot sector at sector 1026048 (offset 0x33300000)

[*] Scan complete. 2 lost partition(s) found.
[+] Rebuilt partition table entry for NTFS starting at
    sector 2048: 000000070000000800000f3d00
[+] Rebuilt partition table entry for FAT32 starting at
    sector 1026048: 00000c0000c0a60f00000f3d00

$ sudo testdisk disk.img    # cross-verification with standard tool
Disk disk.img - 800 MB / 762 MiB
Partition          Start        End    Size
 P NTFS             2048    2050047   1000 MB   [RECOVERED]
 P FAT32         1026048    3074047   1000 MB   [RECOVERED]

$ _
```